

Farhad Zaker

US Permanent Resident (Green Card Holder) – No Sponsorship Required

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PROFESSIONAL SUMMARY

Materials R&D scientist and Mechanical Engineering PhD candidate with a structural engineering background, specializing in high-performance materials development through molecular-scale modeling (MD/DFT) and advanced experimental characterization (SEM, EDS, AFM, FTIR, UV-Vis, mechanical testing). Proven record optimizing complex materials systems—including electrode architectures, thermogalvanic electrolytes, and advanced cementitious composites—evidenced by 26 peer-reviewed publications, 1,500+ citations (h-index 18, i10-index 21), and 2 U.S. provisional patents. Skilled in leading cross-functional research teams, managing research budgets and equipment acquisitions, and serving as Environmental Health & Safety (EHS) liaison for two research facilities.

TECHNICAL SKILLS

Materials Characterization: SEM, EDS, AFM, FTIR, UV-Vis, Zeta Potential Analysis, Surface Profilometry, Galvanic & Thermogalvanic Cell Assembly

Electrochemical & Battery Testing: Electrochemical Impedance Spectroscopy (EIS), Cyclic Voltammetry (CV), Galvanostatic Charge-Discharge (GCD)

Computation & Modeling: Python (SciPy, NumPy, Pandas), MATLAB (Image Processing, GPIB Control), LabVIEW, Molecular Dynamics (LAMMPS, Materials Studio), Abaqus, Origin Pro, statistical data analysis & visualization

Engineering Design & CAD: SolidWorks, AutoCAD, ETABS

EDUCATION

Texas A&M University, College Station, TX

Expected May 2027

PhD in Mechanical Engineering — GPA 3.68

Dissertation: Developing Thermogalvanics in Cementitious Composites for Simultaneous Structural and Heat-Harvesting Capabilities

Sharif University of Technology

Jan 2020

Master of Science in Structural Engineering

University of Science and Technology

Oct 2016

Bachelor of Science in Civil Engineering

EXPERIENCE

Texas A&M University, College Station, TX

Jan 2024 – Present

Graduate Research Assistant — Advisor: Dr. Matthew Powell-Palm

- Led a 2-year effort in hybrid organic–inorganic thermogalvanic electrolyte design, achieving a 36 mV K^{-1} thermopower ($\sim 28\times$ the standard Fe(CN)_6 redox pair) and a $15 \text{ mW K}^{-2} \text{ m}^{-2}$ power factor ($\sim 15\%$ of Carnot).
- Verified a new redox-split heat-harvesting mechanism by leading Cyclic Voltammetry (CV) and spectroscopy (FTIR, UV-Vis) experiments and data analysis.
- Designed SolidWorks models, 3D-printed and fitted with Peltier cells, to build the lab's primary thermogalvanic testing apparatus for electrochemical performance, power curves, and thermal/efficiency metrics.
- Designed and fabricated a custom transient hot-wire (THW) prototype for thermal conductivity measurement, using a GPIB–MATLAB interface to remotely control Keithley multimeter and sourcemeter queries.
- Managed high-value equipment acquisitions (lab-grade multimeters, sourcemeters, vacuum chambers) totaling over \$30,000.
- Served as EHS liaison between the 8-person lab and the Environmental Health & Safety office, leading audits to ensure full OSHA and institutional compliance with zero lost workdays.

Graduate Research Assistant — Advisor: Dr. Amir Asadi

- Provided initial design and managed a team of 4 graduate researchers across 2 labs to build a spray-deposition apparatus for manufacturing nanopatterned LFP cathodes.
- Validated optimal design and patterning via SEM, EDS, and surface profilometry, supporting a \$1M NSF-funded program on multi-material microstructure manufacturing.
- Quantified and optimized the electrochemical–mechanical trade-off, achieving a 14% capacity increase and 4× greater structural cohesion versus slurried lithium iron phosphate.
- Developed and analyzed 12 molecular dynamics (MD) models—three patterns across four materials—in Materials Studio to characterize molecular-scale friction in woven structures (year-long fundamental study).
- Supported an NSF study on sprayed multi-material reinforcement of carbon fiber via SEM characterization, data curation and visualization, and manuscript preparation.

University of Science and Technology

Sep 2015 – Dec 2021

Graduate Researcher — Nanomaterials Research Centre

- Established cross-institution collaboration with UTS and Macquarie University to model MOF-on-MOF cavities, reaching exceptional ion selectivities of 84 (K^+/Li^+) and 80 (Na^+/Li^+).
- Performed high-temperature simulations of bio-oil bonding with sulfur chains in collaboration with Arizona State University, demonstrating that sulfur–carbon formations tripled bitumen elasticity.
- Co-developed MD models of graphene-oxide-nanosheet–cement interactions at the molecular level to track the mechanical properties of microscale composites.
- Conducted rheological (Vicat setting time, flow table), mechanical (compressive, 3-point bending), and durability (chloride permeability) testing on mortar/concrete specimens per ASTM standards, supporting three thesis projects with Dr. Habibnejad.

PUBLICATIONS

Author of 26 peer-reviewed publications in top-tier journals including Advanced Materials, ACS Nano, and Small; 1,500+ citations, h-index 18, i10-index 21 (Google Scholar). Selected publications:

- “A Buffered Hexacyanoferrate Electrolyte for Thermogalvanic Heat Harvesting across Symmetric and Asymmetric Electrode Configurations.” ACS Nano 20.28 (2026): 20176–20187.
- “Thermally-driven chemical heterogeneity produces large thermopower and multiday operability in a thermogalvanic cell.” Journal of Materials Chemistry A 14 (2026): 7695–7705.
- “On a Continuous Aqueous Thermogalvanic Redox Agent with Anomalous Thermopower.” Nano Letters 25.31 (2025): 11986–11992.
- “Patterned Nanostructures on Cathodes: A Pathway to Stronger, High-Energy, High-Power Li-Ion Batteries.” ACS Nano 19.45 (2025): 38970–38980.
- “Multifunctionality through Embedding Patterned Nanostructures in High-Performance Composites.” Advanced Materials 35.32 (2023): 2300948.

Full publication list available on Google Scholar.

PATENTS

2 U.S. provisional patents filed (2026):

- “Thermogalvanic Cementitious Composites.” U.S. Provisional Patent Application (2026).
- “Redox-Split Thermogalvanic Cells, Compositions Therefor, and Methods of Making and Use Thereof.” U.S. Provisional Patent Application No. 63/962,992 (2026).

AWARDS AND ACHIEVEMENTS

Elizabeth & Raymond Marlow '53 Scholarship — Texas A&M University	2025
Full Graduate Research Assistantship — Texas A&M University	2022–2025
Summer Research Grant — Texas A&M University	2022–2024